

Supporting Information

Modelling and experimental studies on phase and chemical equilibria in high pressure methanol synthesis

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1. THERMODYNAMIC PROPERTIES

Table S1. Critical properties, acentric factors, and polarity correction factors for the components involved in methanol synthesis.

	CO	CO ₂	Methanol	H ₂	Water	CH ₄
P _c (MPa)	3.494	7.374	8.097	1.293	22.064	4.599
T _c (K)	132.85	304.12	512.64	32.98	647.14	190.56
ω (-)	0.045	0.225	0.565	-0.217	0.344	0.011
p (-)	0	0	0.2359	0	0.1277	0

The thermodynamic properties are taken from Poling *et al.*¹ and the polarity corrections from Mathias.²

Table S2. Binary interaction parameters (k_{ij}) for the modified SRK EOS.

	CO	CO ₂	Methanol	H ₂	Water	CH ₄
CO	-	0.1164	-0.37	-0.0007	-0.474	0.0204
CO ₂	0.1164	-	0.10	0.1164	0.30	0.0956
Methanol	-0.37	0.10	-	-0.125	-0.075	0.046
H ₂	-0.0007	0.1164	-0.125	-	-0.745	0.001
Water	-0.474	0.30	-0.075	-0.745	-	0.014
CH ₄	0.0204	0.0956	0.046	0.001	0.014	-

The binary interaction parameters for methanol + water, methanol + CO/CO₂/H₂/CH₄, and water + CO/CO₂/H₂/CH₄ are determined from experimental data. The other binary interaction parameters are taken from the Aspen Hysys database.

2. DETERMINATION OF BINARY INTERACTION PARAMETERS AND VALIDATION OF THE MSRK EOS

The liquid phase in methanol synthesis is primarily a binary mixture of water and methanol (see Section 5 of the main article). Furthermore, the vapor phase at equilibrium contains a significant amount of methanol at relevant conditions for high pressure methanol synthesis (20 MPa). To have an accurate description of the methanol synthesis system a good description of the water-methanol VLE is required. The binary interaction coefficient was fitted to literature data obtained at 473 and 523 K.³ The fit is shown in Figure S1 using $k_{ij} = -0.075$. Although only ‘simple’ mixing rules are applied an accurate fit is obtained.

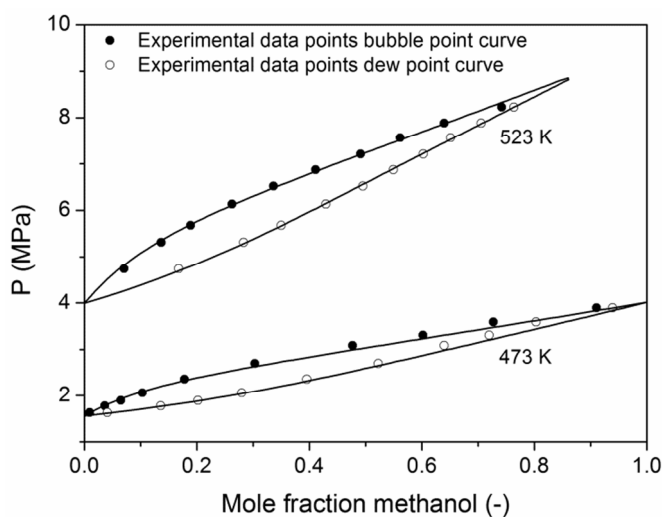


Figure S1. VLE diagram of water and methanol. Symbols: experimental data,³ lines: model predictions using $k_{ij} = -0.075$.

Binary interaction coefficients for VLE of methanol and H_2 , CO , CO_2 , and CH_4 respectively were obtained by fitting the MSRK EOS to VLE data from Brunner *et al.*⁴. The experimental data points for each binary mixture were selected at temperatures the closest to methanol synthesis conditions, see

Figure S2 and Table S2 for the binary interaction coefficients derived. Also here, the behavior of the VLE is described qualitatively correct, but the discrepancies between the experimental data and the model predictions are somewhat larger than for the water-methanol VLE. The H_2 , CO , and CH_4 dew point curves are accurately predicted, while the predictions of the bubble point curves are less accurate. For CO_2 the number of data points is limited. The qualitative description of the system is good, but both the prediction of the dew point curve and the bubble point curve differ from the experimental values.

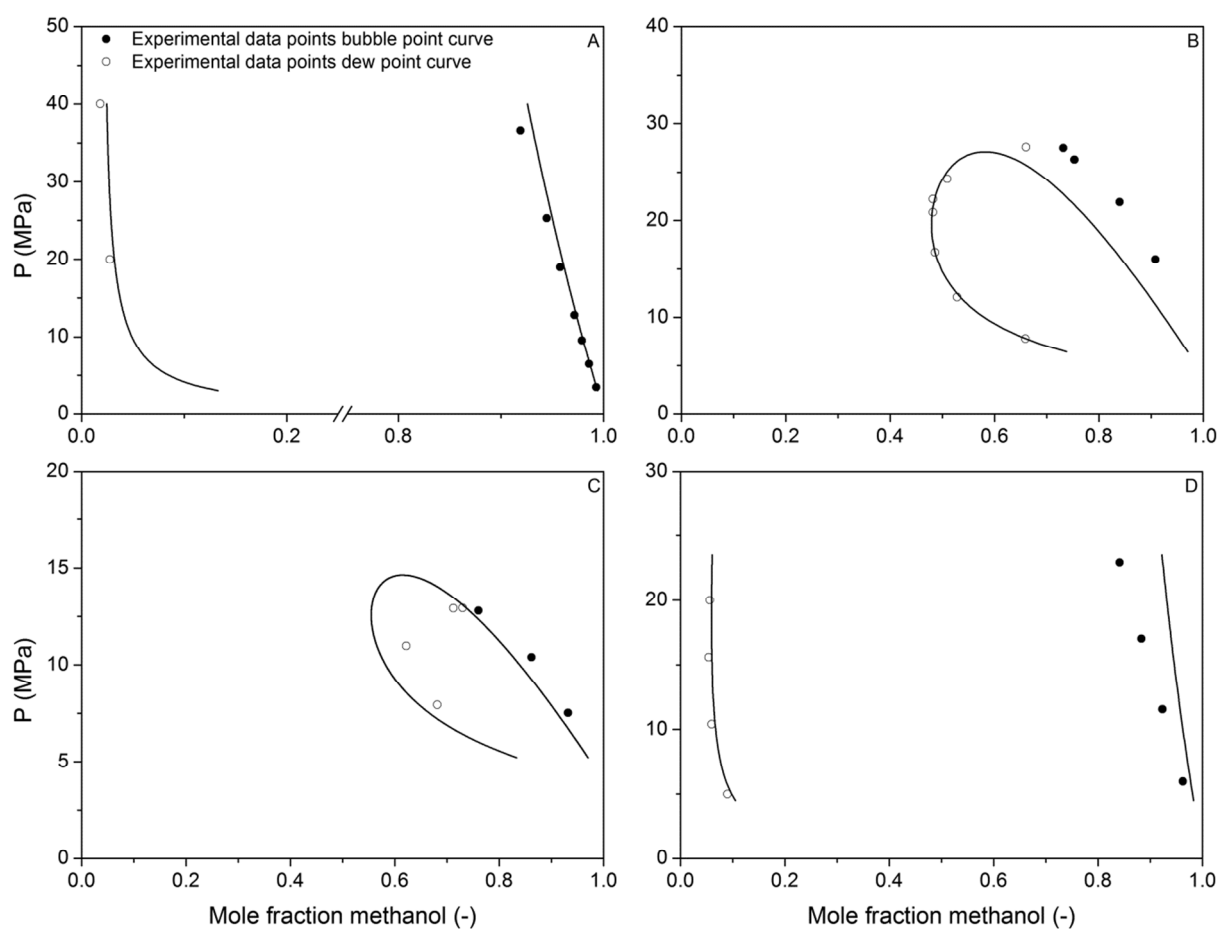


Figure S2. VLE diagrams of methanol- H_2 at 373 K (A, $k_{ij} = -0.125$), methanol- CO at 473 K (B, $k_{ij} = -0.37$), methanol- CO_2 at 473 K (C, $k_{ij} = 0.10$), methanol- CH_4 at 373 K (D, $k_{ij} = 0.046$). Symbols: experimental data,⁴ lines: model predictions.

In Figure S3 the solubility of H₂, CO, and CH₄ in water and the VLE of CO₂ and water are given. The experimental solubility of H₂ was measured by Kling and Maurer,⁵ the solubility of CO by Dake and Chaudari,⁶ and the solubility of CH₄ by Culberson and McKetta.⁷ For the system of CO₂ and water, VLE data were taken from Müller *et al.*⁸

The predictions of the solubility data and the VLE of water-CO₂ are very accurate. The solubility predictions for CO differ slightly from the experimental values, but the number of data points is limited. As for the methanol binary systems (Figure S1 and S2) the binary interaction parameters were derived from these data (Figure S3). The binary interaction parameters for the other binary pairs are taken from the Aspen Hysys database for the SRK EOS and can be found in Table S2.

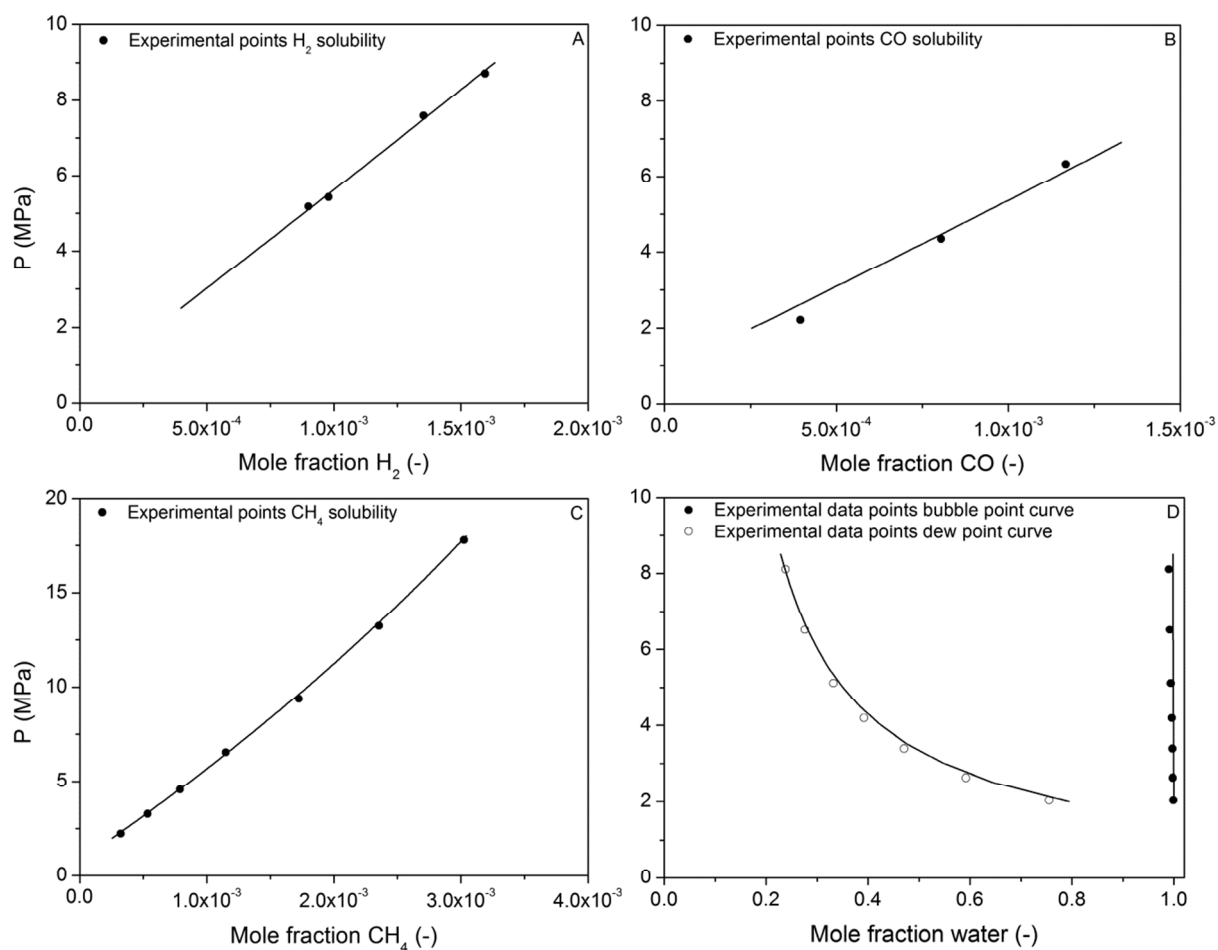


Figure S3. Solubility of H₂ in water at 423 K (A, $k_{ij} = -0.745$),⁵ solubility of CO in water at 448 K (B, $k_{ij} = -0.474$),⁶ solubility of CH₄ in water at 444 K (C, $k_{ij} = 0.014$),⁷ VLE diagram of water-CO₂ at 473 K (D, $k_{ij} = 0.30$).⁸ Symbols: experimental data, lines: model predictions.

3. COMPARISON OF THIS WORK WITH LITERATURE

Table S3. Overview of modelled equilibrium compositions for the vapor and liquid phase in high pressure methanol synthesis. P = 30 MPa, T = 473 K, feed gas: H₂/CO/CO₂/CH₄ = 74/15/8/3 vol%.

	Feed	Castier et al. ⁹	Gupta et al. ¹⁰	Stateva and Wakeham ¹¹	Jalali and Seader ¹²	Avami and Sabooshi ¹³	This work
Vapor phase							
CO ^a	15	6·10 ⁻³	0.01	1.3·10 ⁻³	6.2·10 ⁻³	Traces	8.6·10 ⁻³
CO ₂ ^a	8	0.05	0.04	Traces	0.05	Traces	0.06
Methanol ^a	0	20.53	20.40	21.20	20.42	16.55	20.23
H ₂ ^a	74	65.89	66.93	64.93	66.02	70.91	65.51
Water ^a	0	4.73	3.74	4.64	4.71	3.53	5.06
CH ₄ ^a	3	8.75	9.15	9.23	8.79	9.02	9.13
Phase frac.	-	0.5112	0.5258	0.4968	0.4863	0.4825	0.4848
Liquid phase							
CO ^a	-	Traces	Traces	Traces	Traces	Traces	1.8·10 ⁻³
CO ₂ ^a	-	0.02	0.01	Traces	0.02	Traces	0.02
Methanol ^a	-	63.54	67.09	63.71	63.43	66.87	63.43
H ₂ ^a	-	9.62	4.20	9.48	9.76	5.47	10.49
Water ^a	-	24.36	27.32	24.88	24.29	25.33	23.87
CH ₄ ^a	-	2.46	1.56	1.93	2.48	2.33	2.19
Phase frac.	-	0.4888	0.4742	0.5032	0.5137	0.5175	0.5152

^aAll concentration are expressed in mol%.

4. LITERATURE

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